

Paper 2 - Set 03 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

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1	E	Indices and surds
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3	F	Differentiation
4	A	Identifying errors in proofs
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Worked solutions

Question 1 IND-13 · Indices and surds · spec MM1.1

Simplify $\left(\frac{a^{\frac{1}{2}}b^{-2}}{a^{-1}b^{\frac{1}{2}}}\right)^4$.

Answer: E

Inside the bracket: $a^{\frac{1}{2}+1}b^{-2-\frac{1}{2}} = a^{\frac{3}{2}}b^{-\frac{5}{2}}$. Raise to the 4th power: a^6b^{-10} .

Full method - how and why

Index laws with two variables. Simplify inside the bracket first, then apply the outside power to each index.

1 Simplify the a 's inside: $\frac{a^{\frac{1}{2}}}{a^{-1}} = a^{\frac{1}{2}-(-1)} = a^{\frac{3}{2}}$.

why: Dividing subtracts indices, and subtracting a negative adds - the commonest slip in the question.

2 Simplify the b 's: $\frac{b^{-2}}{b^{\frac{1}{2}}} = b^{-2-1/2} = b^{-5/2}$.

why: Same rule; keep the two variables in separate columns so signs do not get mixed up.

3 Raise the bracket to the 4th power: $(a^{\frac{3}{2}}b^{-5/2})^4 = a^6b^{-10}$.

why: $(a^m)^n = a^{mn}$ applies to each base separately: $\frac{3}{2} \times 4 = 6$ and $-\frac{5}{2} \times 4 = -10$.

Where it goes wrong

- Getting $a^{-1/2}$ at step 1 by subtracting the wrong way round, leading to $a^{-2}b^{10}$ - which is offered.
- Applying the power 4 to only the first factor.

Takeaway Simplify inside the bracket to a single power of each base before applying the outer index.

Question 2 LOG-09 · Exponentials and logarithms · spec MM5.2

Find the value of $\frac{\log_5 27}{\log_5 3}$.

Answer: D

$\log_5 27 = \log_5 3^3 = 3\log_5 3$, so the quotient is 3.

Full method - how and why

A quotient of two logs with the same base. Rewriting the numerator's argument as a power makes it cancel.

1 Write $27 = 3^3$, so $\log_5 27 = \log_5 3^3 = 3\log_5 3$.

why: The power law applies to the argument, whatever the base of the log.

2 Form the quotient: $\frac{3\log_5 3}{\log_5 3} = 3$.

why: The common factor $\log_5 3$ cancels (it is non-zero).

Where it goes wrong

- Subtracting to get $\log_5 24$ - a quotient of logs is not the log of a quotient.
- Dividing the arguments to get $\frac{27}{3} = 9$ in log form.

Takeaway $\frac{\log_c a}{\log_c b} = \log_b a$: quotients of logs change the base, they do not divide the arguments.

Question 3 DIF-05 · Differentiation · spec MM6.3

For what range of values of x is $f(x) = 2x^3 - 15x^2 + 24x + 1$ decreasing?

Answer: F

$$f'(x) = 6x^2 - 30x + 24 = 6(x-1)(x-4) < 0 \text{ for } 1 < x < 4.$$

Full method - how and why

'Decreasing' is the condition $f'(x) < 0$ - a quadratic inequality once the derivative is found.

1 Differentiate: $f'(x) = 6x^2 - 30x + 24$.

why: Term by term.

2 Factorise: $6(x^2 - 5x + 4) = 6(x-1)(x-4)$.

why: Taking out the 6 makes the roots visible.

3 Solve $f'(x) < 0$: the upward parabola is negative between its roots, so $1 < x < 4$.

why: The gradient is negative exactly where the function falls.

Where it goes wrong

- Solving $f'(x) > 0$ and answering $x < 1$ or $x > 4$ - which is offered.
- Forgetting the factor 6 when factorising, giving the wrong roots.

Takeaway Decreasing $\Leftrightarrow f' < 0$; find the roots of f' and use the shape of its graph.

Question 4 ERR-10 · Identifying errors in proofs · spec Err2

Claim: if $a > b$ then $\frac{1}{a} < \frac{1}{b}$.

I Suppose $a > b$.

II Dividing both sides by ab : $\frac{1}{b} > \frac{1}{a}$.

III So $\frac{1}{a} < \frac{1}{b}$.

Which of the following is correct?

Answer: A

$a = 1, b = -1$: $a > b$ but $\frac{1}{a} = 1 > -1 = \frac{1}{b}$. Dividing by ab without knowing its sign is invalid (and $b = 0$ would be division by zero).

Full method - how and why

The same unknown-sign trap as ERR-05, now with division, plus a false claim.

1 Test the claim: $a = 1, b = -1$. Then $a > b$ but $\frac{1}{a} = 1$ and $\frac{1}{b} = -1$, so $\frac{1}{a} > \frac{1}{b}$. **Claim false.**

why: Opposite signs are the natural test for a reciprocal claim.

2 Line I is just the supposition - fine.

why: No deduction yet.

3 Line II divides by ab and keeps the direction. That needs $ab > 0$, which is not given; ab could be negative (as in the counterexample) or zero. **First error.**

why: Dividing by a negative reverses the inequality, and dividing by zero is undefined.

4 The claim is true if a and b have the same sign - which is the condition that was silently assumed.

why: Naming the missing hypothesis is the cleanest way to describe this kind of error.

Where it goes wrong

- Accepting the argument because it 'works for positive numbers' - the claim is stated for all reals.
- Blaming line III, which is a trivial restatement of line II.

Takeaway Reciprocals reverse order only within a fixed sign: for $a, b > 0$ (or both negative), $a > b \Rightarrow \frac{1}{a} < \frac{1}{b}$.

Question 5 LGC-19 · The logic of arguments · spec Arg2

Let f be a function defined for all real numbers. P : ' $f'(a) = 0$ '; Q : ' f has a stationary point at $x = a$ '. Which one of the following is true?

Answer: A

A stationary point is by definition a point where the derivative is zero, so P and Q are equivalent.

Full method - how and why

Sometimes the two statements are the same by **definition** - recognise that and both directions are immediate.

- 1 Recall the definition: f has a stationary point at $x = a$ precisely when $f'(a) = 0$.

why: That is what 'stationary' means - the rate of change is zero.

- 2 So $P \Rightarrow Q$ and $Q \Rightarrow P$ both hold trivially.

why: Definitions are biconditional.

- 3 Hence P is necessary and sufficient for Q .

why: Match to the option list.

Where it goes wrong

- Confusing this with 'turning point': $f'(a) = 0$ does **not** guarantee a maximum or minimum ($y = x^3$ at $x = 0$). But the question says stationary point, not turning point.
- Worrying about functions that are not differentiable - the question states f is defined and $f'(a)$ is referred to, so differentiability is assumed.

Takeaway A definition gives both implications at once; but stationary $\neq$ turning - the extra condition is a **sign change** in f' .

Question 6 ERR-16 · Identifying errors in proofs · spec Err2

A student argues: 'Every prime number is odd, because if it were even it would be divisible by 2 and so not prime.' Which of the following is correct?

Answer: A

A number divisible by 2 is not prime unless it equals 2. The conclusion 'every prime is odd' is false.

Full method - how and why

An argument that proves 'almost' the right thing. Find the single case it mishandles.

- 1 The reasoning is: even $\Rightarrow$ divisible by 2 $\Rightarrow$ not prime.

why: Trace the chain and test each link.

- 2 The second link fails for one number: 2 is divisible by 2 and **is** prime, because divisibility by itself is allowed.

why: A prime has exactly two divisors, 1 and itself; for 2 those are 1 and 2.

- 3 So the conclusion 'every prime is odd' is false - 2 is an even prime.

why: A single counterexample refutes it.

- 4 The correct statement is 'every prime **greater than 2** is odd', and the argument does establish that.

why: Identifying the corrected claim shows precisely what the flaw was.

Where it goes wrong

- Objecting that some odd numbers are not prime - true, but irrelevant; the claim is not the converse.
- Objecting via 1 or 0 - neither is a counterexample to 'every prime is odd'.

Takeaway Check whether an argument's chain breaks at a boundary case. 2 is the even prime that breaks this one.

Question 7 X2-24 · The logic of arguments · spec Arg2

For a real number x , P : ' $x > 0$ '. Q : ' $x^3 > 0$ '. Which one of the following is true?

Answer: A

Cubing preserves sign, so $x > 0 \Leftrightarrow x^3 > 0$.

Full method - how and why

Cubing preserves sign in both directions, so this is an equivalence - unlike squaring.

1 If $x > 0$ then $x^3 = x \cdot x \cdot x > 0$: **sufficient**.

why: A product of three positives is positive.

2 Conversely, if $x \leq 0$ then $x^3 \leq 0$ (negative cubed is negative, and $0^3 = 0$). So $x^3 > 0$ forces $x > 0$: **necessary**.

why: Proved by contrapositive - check the negative and zero cases rather than hunting for a counterexample.

3 So P is necessary and sufficient.

why: Cubing is a strictly increasing, sign-preserving operation.

Where it goes wrong

- Importing the squaring counterexample ($x = -2$) - for cubes $(-2)^3 = -8 < 0$, so it does not apply.
- Answering 'sufficient but not necessary' by pattern-matching to x^2 questions.

Takeaway Odd powers preserve sign and order; even powers do not. $x > 0 \Leftrightarrow x^3 > 0$.

Question 8 ERR-01 · Identifying errors in proofs · spec Err2

A student attempts to solve the equation $\sqrt{x+3} = x-3$ for real x .

I Squaring both sides: $x+3 = (x-3)^2 = x^2 - 6x + 9$.

II So $x^2 - 7x + 6 = 0$.

III So $(x-1)(x-6) = 0$.

IV Hence the solutions are $x = 1$ and $x = 6$.

Which of the following best describes the solution?

Answer: A

Squaring can introduce extraneous solutions, so line IV should check them: $x = 1$ gives $\sqrt{4} = 2 \neq -2$. Only $x = 6$ works. Lines I-III are valid deductions.

Full method - how and why

Squaring an equation is not reversible: it can create solutions that do not satisfy the original. The error is the **missing check**, not the algebra.

1 Check line I: squaring both sides of $\sqrt{x+3} = x-3$ gives $x+3 = x^2 - 6x + 9$. Valid.

why: $A = B \Rightarrow A^2 = B^2$ is always a legitimate deduction - it is the converse that fails.

2 Check lines II and III: $x^2 - 7x + 6 = 0$ factorises as $(x-1)(x-6) = 0$. Valid.

why: $-1 \times -6 = 6$ and $-1 - 6 = -7$ ✓.

3 Line IV asserts both are solutions. Substitute $x = 1$ into the **original**: $\sqrt{4} = 2$, but $x-3 = -2$. $2 \neq -2$, so $x = 1$ is extraneous.

why: A square root is non-negative, so the right-hand side must be ≥ 0 , forcing $x \geq 3$.

4 Substitute $x = 6$: $\sqrt{9} = 3$ and $6-3 = 3$ ✓. So only $x = 6$ is a solution, and the first error is line IV.

why: Lines I-III produce **candidates**; the check is a required final step.

Where it goes wrong

- Blaming line I - squaring is a valid implication, just not an equivalence.
- Declaring the solution correct because the factorising is right; correct algebra can still give a wrong answer set.
- Checking the candidates in the squared equation rather than the original - both pass there, which proves nothing.

Takeaway After squaring, always substitute back into the **original** equation; squaring can only add solutions, never lose them.

Question 9 X2-22 · Exponentials and logarithms · spec MM5.3

Consider the statement: ****for all real x , $2^x > x^2$.**** Which of the following is a counterexample?

Answer: B

$2^3 = 8 < 9 = 3^2$, so $x = 3$ breaks the claim. The other options satisfy it: $2 > 1$, $32 > 25$, $1 > 0$, $1024 > 100$.

Full method - how and why

Exponential versus quadratic growth: the exponential eventually wins, but there is a window where it loses. Test values inside that window.

- 1 Check the candidates: $x = 0$ gives $1 > 0$ ✓; $x = 1$ gives $2 > 1$ ✓; $x = 5$ gives $32 > 25$ ✓; $x = 10$ gives $1024 > 100$ ✓.

why: Evaluate both sides for each - no cleverness needed.

- 2 $x = 3$: $2^3 = 8$ and $3^2 = 9$. Since $8 < 9$, the claim fails. **Counterexample.**

why: Hypothesis (x real) satisfied, conclusion false.

- 3 Why here? 2^x and x^2 cross at $x = 2$ (both 4) and again near $x = 4$ (both 16), so the quadratic is ahead between them.

why: Knowing where the curves cross tells you exactly where to hunt.

Where it goes wrong

- Testing only large x , where the exponential dominates and the claim holds.
- Overlooking the window: the two sides are equal at $x = 2$ and $x = 4$, so the quadratic leads strictly between them - that is where a counterexample must lie.
- Assuming exponentials always beat polynomials - true only eventually.

Takeaway 2^x and x^2 meet at $x = 2$ and $x = 4$; between them the quadratic is larger. 'Eventually bigger' is not 'always bigger'.

Question 10 X2-19 · Probability · spec M7.7

Events A and B have $P(A) > 0$ and $P(B) > 0$. P : ' A and B are mutually exclusive'. Q : ' A and B are not independent'. Which one of the following is true?

Answer: C

Mutually exclusive gives $P(A \text{ and } B) = 0 \neq P(A)P(B)$, so they cannot be independent: P is sufficient. But two overlapping events can also fail to be independent, so P is not necessary.

Full method - how and why

Relating mutual exclusivity to independence. One implication holds; the converse has easy counterexamples.

- 1 If A, B are mutually exclusive, $P(A \cap B) = 0$. But $P(A)P(B) > 0$ since both probabilities are positive.

why: The positivity hypothesis is what makes the argument work.

- 2 So $P(A \cap B) \neq P(A)P(B)$: they are **not** independent. $P \Rightarrow Q$: **sufficient**.

why: Independence is exactly the equality that fails here.

- 3 Necessity: find dependent events that are **not** mutually exclusive. Roll a die with A : 'even' and B : 'at most 3'.

Then $P(A \cap B) = P(\{2\}) = \frac{1}{6}$, but $P(A)P(B) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

why: Unequal, so they are dependent - yet they overlap, so they are not mutually exclusive.

- 4 So P is sufficient but not necessary.

why: Match to the list.

Where it goes wrong

- Treating mutually exclusive and independent as opposites in a way that makes them necessary for each other - dependence has many other causes.
- Forgetting the hypothesis $P(A), P(B) > 0$; with a zero probability, events can be both mutually exclusive and independent.

Takeaway For events of positive probability, mutually exclusive $\Rightarrow$ dependent, but not conversely.

Question 11 X2-01 · Quadratic functions · spec MM1.3

p and q are real numbers. Which of the following statements is/are true?

- I If $p^2 < 4q$ then $x^2 + px + q > 0$ for all real x .
- II If $x^2 + px + q > 0$ for all real x then $q > 0$.
- III If $q < 0$ then $x^2 + px + q = 0$ has two real roots of opposite sign.

Answer: H

I: negative discriminant and positive leading coefficient. II: put $x = 0$. III: the product of the roots is $q < 0$ (and the discriminant $p^2 - 4q > 0$), so the roots are real with opposite signs.

Full method - how and why

Three claims about a quadratic. Each is settled by the discriminant, by a well-chosen substitution, or by the root-product relationship.

- 1 I: $p^2 < 4q$ means the discriminant $p^2 - 4q < 0$, so $x^2 + px + q$ has no real roots; with a positive x^2 coefficient the graph lies entirely above the axis. **True.**
why: No roots plus an upward parabola means always positive - the graph never reaches the axis.
- 2 II: if $x^2 + px + q > 0$ for **all** x , then in particular at $x = 0$: $q > 0$. **True.**
why: Substituting a convenient value is the quickest way to extract information from a universal statement.
- 3 III: the product of the roots is q . If $q < 0$ then $p^2 - 4q > p^2 \geq 0$, so the roots are real and distinct, and their product being negative makes them opposite in sign. **True.**
why: For $x^2 + px + q$ the sum of roots is $-p$ and the product is q .
- 4 So all three.
why: Match to the list.

Where it goes wrong

- Reading I's condition as $p^2 > 4q$ - that would give two real roots and the opposite conclusion.
- Forgetting to check in III that the roots are **real**; a negative product alone would be meaningless for complex roots.
- Trying to prove II with a general argument when $x = 0$ settles it in one line.

Takeaway Discriminant controls the number of roots; $x = 0$ extracts the constant term; sum and product of roots are $-b/a$ and c/a .

Question 12 X2-11 · Functions and transformations · spec MM8.2

Let $f(x) = x^2$ for all real x , and let g be the function whose graph is obtained by translating the graph of $y = f(x)$ by 2 units in the positive x -direction and then reflecting in the y -axis. Which of the following statements is/are true?

- I** $g(x) = (x + 2)^2$
- II** $g(x) = (x - 2)^2$
- III** The graph of g has its vertex at $(-2, 0)$

Answer: F

Translation gives $(x - 2)^2$; reflecting in the y -axis replaces x by $-x$: $(-x - 2)^2 = (x + 2)^2$, with vertex at $(-2, 0)$.

Full method - how and why

Composite transformations, applied in the stated order. Each transformation acts on the function you currently have.

- 1** Translate $y = x^2$ by 2 in the positive x -direction: replace x by $x - 2$, giving $y = (x - 2)^2$.
why: A shift **right** subtracts inside the bracket - counter-intuitive but standard.
- 2** Now reflect in the y -axis: replace x by $-x$, giving $y = (-x - 2)^2$.
why: Reflection in the y -axis is the substitution $x \rightarrow -x$.
- 3** Simplify: $(-x - 2)^2 = (-(x + 2))^2 = (x + 2)^2$. So $g(x) = (x + 2)^2$ - **I is true, II is false.**
why: Squaring removes the overall minus sign.
- 4** III: $(x + 2)^2$ has its vertex where $x + 2 = 0$, i.e. at $(-2, 0)$. **True.**
why: Consistent: the vertex moved right to $(2, 0)$, then reflected to $(-2, 0)$.
- 5** So I and III.
why: Match to the list.

Where it goes wrong

- Doing the transformations in the wrong order - reflecting first then translating gives $(x - 2)^2$ instead.
- Shifting right by adding 2 inside the bracket.
- Forgetting to simplify $(-x - 2)^2$, and so failing to recognise it as $(x + 2)^2$.

Takeaway Right by a : $x \rightarrow x - a$. Reflect in the y -axis: $x \rightarrow -x$. Apply them in the order given, tracking the vertex as a check.

Question 13 RSN-13 · Chains of reasoning · spec Prf5

p and q are prime numbers with $p + q = 45$. Find pq .

Answer: D

An odd sum of two primes needs one of them to be 2 (the only even prime): $q = 43$, which is prime. $pq = 86$.

Full method - how and why

Parity applied to primes: the sum of two primes is odd only if one of them is 2.

- 1 45 is odd, so one prime is even and one odd.
why: odd + odd = even, even + even = even; only odd + even gives an odd total.
- 2 The only even prime is 2, so $p = 2$ and $q = 43$.
why: 2 is the unique even prime - every other even number has 2 as a proper factor.
- 3 Check 43 is prime: not divisible by 2, 3, 5 and $7^2 = 49 > 43$, so yes.
why: Test primes up to the square root only.
- 4 $pq = 2 \times 43 = 86$.
why: The question asks for the product, not the primes.

Where it goes wrong

- Trying odd pairs like $2 + 43$ versus $5 + 40$ at random; 40 is not prime and no odd pair can work at all.
- Answering 43 or 45 instead of the product.

Takeaway An odd sum of two primes forces one of them to be 2. Parity settles most prime puzzles instantly.

Question 14 RSN-11 · Chains of reasoning · spec Prf5

The product of three consecutive positive integers is N , and the middle one of the three integers is m .

Which of the following statements must be true?

- I N is divisible by 6.
- II N is divisible by 4.
- III $N + m$ is a perfect cube.

Answer: F

Among three consecutive integers there is a multiple of 2 and a multiple of 3 (I). II fails for $1 \cdot 2 \cdot 3 = 6$. III:
 $N + m = (m - 1)m(m + 1) + m = m^3 - m + m = m^3$.

Full method - how and why

Three consecutive integers: automatic divisibility by 2 and 3, but not by 4; plus a neat algebraic identity.

- 1 Write the three as $m - 1, m, m + 1$, so $N = (m - 1)m(m + 1)$.
why: Centring on the middle term is what makes III come out.
- 2 I: among any three consecutive integers there is at least one multiple of 2 and one multiple of 3, so $6 \mid N$. **True.**
why: Every second integer is even and every third is a multiple of 3.
- 3 II: test the smallest case $1 \times 2 \times 3 = 6$, which is not divisible by 4. **False.**
why: Three consecutive integers contain only one or two evens; when only one, it may contribute a single factor of 2.
- 4 III: $N + m = (m^2 - 1)m + m = m^3 - m + m = m^3$ - a perfect cube. **True.**
why: $(m - 1)(m + 1) = m^2 - 1$ is the difference of squares; the $-m$ and $+m$ cancel.
- 5 So I and III.
why: Match to the list.

Where it goes wrong

- Accepting II by testing $2 \times 3 \times 4 = 24$ only - one confirming case is not enough.
- Expanding N the long way instead of using $(m - 1)(m + 1) = m^2 - 1$.

Takeaway k consecutive integers are always divisible by $k!$ - here $3! = 6$, but not necessarily by 4.

Question 15 X2-18 · Number, units and estimation · spec M2.3

n is a positive integer greater than 1. Which of the following statements is/are true?

- I If n is not prime, then n has a prime factor at most $\sqrt{n}$.
- II If n has no prime factor at most $\sqrt{n}$, then n is prime.
- III If n is prime, then n has no factor between 1 and n exclusive.

Answer: H

If $n = ab$ with $1 < a \leq b$ then $a \leq \sqrt{n}$ and a has a prime factor $\leq \sqrt{n}$ (I). II is the contrapositive of I. III is the definition of prime.

Full method - how and why

Three number-theory statements about primality testing, two of which are contrapositives of each other.

- 1 I: if n is composite write $n = ab$ with $1 < a \leq b$. Then $a^2 \leq ab = n$, so $a \leq \sqrt{n}$.
why: The smaller factor of a pair cannot exceed the square root, or the product would be too big.
- 2 $a > 1$ has a prime factor, which is at most $a \leq \sqrt{n}$. So n has a prime factor $\leq \sqrt{n}$. **True.**
why: Every integer greater than 1 has a prime factor, and it cannot exceed the number itself.
- 3 II: 'no prime factor $\leq \sqrt{n} \Rightarrow n$ prime' is exactly the contrapositive of I, so it is **True** automatically.
why: Contrapositives share truth values - no separate work needed.
- 4 III: 'if n is prime it has no factor strictly between 1 and n ' is the definition of prime. **True.**
why: A prime's only positive factors are 1 and itself.
- 5 So all three.
why: Match to the list.

Where it goes wrong

- Treating II as the converse of I and rejecting it; it is the contrapositive (negate and swap).
- Wording III as 'no factors at all', which is wrong: 1 and n are factors.

Takeaway To test n for primality, trial-divide by primes up to $\sqrt{n}$ - and I/II are the two readings of why that works.

Question 16 PRF-05 · Mathematical proof · spec Prf3

The sums 1 , $1 + 3$, $1 + 3 + 5$, $1 + 3 + 5 + 7$ are 1 , 4 , 9 , 16 . A student conjectures that the sum of the first n odd numbers is n^2 . Which of the following is a valid justification?

Answer: D

A applies the arithmetic-series formula to all n . (C also gives a correct calculation but B, D and E are not justifications.)

Full method - how and why

Distinguishing a justification from a pattern-spot. The valid answer must establish the formula for all n .

- 1** The first n odd numbers are $1, 3, 5, \dots$ - an arithmetic series with $a = 1, d = 2$.

why: Recognising the structure is what lets a general formula apply.

- 2** Apply $S_n = \frac{n}{2}(2a + (n - 1)d) = \frac{n}{2}(2 + 2(n - 1))$.

why: The standard arithmetic-series sum, valid for every n .

- 3** Simplify: $\frac{n}{2} \times 2n = n^2$.

why: $2 + 2n - 2 = 2n$.

- 4** That holds for all n , so it justifies the conjecture.

why: By contrast, 'the pattern continues with 25' checks one more case, and 'four cases is enough' explicitly is not a proof.

Where it goes wrong

- Accepting 'four cases have been checked' - the whole point of this topic is that it is not enough.
- Accepting ' n^2 is odd when n is odd' - a true but completely irrelevant statement.
- Being drawn to the even-numbers argument; its calculation happens to work out, but it is presented with an unjustified aside, whereas the series argument is clean and standard.

Takeaway A justification must apply to every n ; extending a pattern one more term never does.

Question 17 ALG-08 · GCSE algebra · spec M4.19

The n th term of a sequence is $\frac{n^2 + 1}{2n}$. Which of the following statements is/are true?

- I The 5th term is 2.6.
- II Every term after the first is greater than the previous term.
- III Some term of the sequence equals 10.

Answer: E

I: $\frac{26}{10} = 2.6$. II: $u_n = \frac{n}{2} + \frac{1}{2n}$; $u_1 = 1$, $u_2 = 1.25$, and for $n \geq 1$ the increase $\frac{1}{2} - \frac{1}{2n(n+1)} > 0$, so increasing. III: $n^2 - 20n + 1 = 0$ has no integer solutions (discriminant 396 is not a square), so false.

Full method - how and why

Three claims about a sequence. Splitting the formula makes the behaviour transparent.

1 I: $u_5 = \frac{25 + 1}{10} = \frac{26}{10} = 2.6$. **True.**

why: Direct substitution.

2 Split the formula: $u_n = \frac{n^2 + 1}{2n} = \frac{n}{2} + \frac{1}{2n}$.

why: Splitting shows one growing part and one shrinking part - much easier to reason about.

3 II: $u_{n+1} - u_n = \frac{1}{2} - \frac{1}{2n(n+1)}$, which is positive for all $n \geq 1$ since $\frac{1}{2n(n+1)} \leq \frac{1}{4}$. **True.**

why: The $\frac{n}{2}$ term grows by $\frac{1}{2}$ each step, while the $\frac{1}{2n}$ term shrinks by less than that.

4 III: solve $\frac{n^2 + 1}{2n} = 10 \Rightarrow n^2 - 20n + 1 = 0$, whose discriminant $400 - 4 = 396$ is not a perfect square, so n is not an integer. **False.**

why: Terms exist only for whole-number n , so a non-integer solution means no term equals 10.

5 I and II only.

why: Match to the options.

Where it goes wrong

- Accepting III because the equation has real solutions - they must be **positive integers**.
- Testing II on only the first two terms.

Takeaway Split a rational formula into simpler pieces; and remember sequence indices must be whole numbers.

Question 18 GRA-15 · Graphs of functions · spec MM8.6

The cubic $y = x^3 + ax + b$ has three distinct real roots. Which of the following statements must be true?

- I $a < 0$
- II $b \neq 0$
- III The curve has two turning points.

Answer: F

Three distinct real roots require two turning points (III), which needs $y' = 3x^2 + a$ to have two roots, so $a < 0$ (I). II is false: $x^3 - x$ has roots $0, \pm 1$ and $b = 0$.

Full method - how and why

Conditions for a cubic to have three distinct roots. Two claims follow from the shape; the third needs a counterexample.

- 1 III: three distinct real roots force the curve to turn twice (up, down, up), so it must have two turning points.

True.

why: Between consecutive roots the curve must turn around.

- 2 I: $y' = 3x^2 + a$ has two real roots only when $a < 0$ (so that $x^2 = -\frac{a}{3}$ has solutions). **True.**

why: This follows directly from III: two turning points require y' to have two roots.

- 3 II: test $y = x^3 - x = x(x - 1)(x + 1)$: three distinct roots $0, \pm 1$, and here $b = 0$. **False.**

why: One counterexample is enough; $b = 0$ simply means the origin is a root.

- 4 I and III only.

why: Match to the options.

Where it goes wrong

- Assuming $b \neq 0$ because 'otherwise the cubic factorises' - it does, and that is fine.
- Failing to link I to III and guessing the sign of a .

Takeaway Three distinct roots $\Leftrightarrow$ two turning points with values of opposite sign; that forces $a < 0$ but says nothing about b .

Question 19 PRF-15 · Mathematical proof · spec Prf3

Let $u_n = n^2 - n + 41$. Which of the following statements is/are true?

- I u_n is prime for $n = 1, 2, 3, 4, 5$.
- II u_n is prime for all positive integers n .
- III u_{41} is divisible by 41.

Answer: F

$u_1 = 41, u_2 = 43, u_3 = 47, u_4 = 53, u_5 = 61$: all prime (I). $u_{41} = 41^2 - 41 + 41 = 41^2$, divisible by 41 (III) and not prime, so II is false - small cases do not prove a general claim.

Full method - how and why

The classic warning about pattern-spotting: a formula can produce primes for many values and still fail.

1 I: compute $u_1 = 41, u_2 = 4 - 2 + 41 = 43, u_3 = 9 - 3 + 41 = 47, u_4 = 16 - 4 + 41 = 53, u_5 = 25 - 5 + 41 = 61$. All prime. **True.**

why: Straight substitution; all five are primes (none divisible by 2, 3, 5 and each is below 64 or checkable quickly).

2 III: $u_{41} = 41^2 - 41 + 41 = 41^2$. Divisible by 41. **True.**

why: The -41 and $+41$ cancel exactly - which is why $n = 41$ is the natural place to look for failure.

3 II: since $u_{41} = 41^2 = 1681$ has factor 41, it is not prime. **False.**

why: One counterexample kills the universal claim, however many earlier cases worked.

4 So I and III.

why: Match to the list.

Where it goes wrong

- Accepting II because the first several values are prime - this formula is famous precisely because it works for $n = 1$ to 40 and then fails.
- Trying to check u_{41} by multiplying out 1681 instead of spotting the cancellation.

Takeaway Many confirming cases never prove a universal claim; and when hunting for failure, look for the value that makes terms cancel.

Question 20 RSN-18 · Chains of reasoning · spec Prf5

How many ordered pairs of positive integers (m, n) satisfy $\frac{1}{m} + \frac{1}{n} = \frac{1}{4}$?

Answer: F

$4n + 4m = mn \Rightarrow (m - 4)(n - 4) = 16$. Positive factor pairs of 16 as ordered pairs:
 $(1, 16), (2, 8), (4, 4), (8, 2), (16, 1)$: five. (Negative factor pairs would make m or n non-positive.)

Full method - how and why

Clear the fractions, then use the add-a-constant factorising trick and count **ordered** pairs.

1 Multiply $\frac{1}{m} + \frac{1}{n} = \frac{1}{4}$ by $4mn$: $4n + 4m = mn$.

why: Clearing denominators turns it into a polynomial equation in integers.

2 Rearrange and factorise: $mn - 4m - 4n + 16 = 16$, so $(m - 4)(n - 4) = 16$.

why: Add $(-4)(-4) = 16$ to both sides to complete the rectangle.

3 m, n positive: if $m - 4$ were negative then $m \in \{1, 2, 3\}$ and $n - 4 = \frac{16}{m-4}$ would make n negative. So both factors are **positive**.

why: Check the negative branch explicitly rather than assuming; e.g. $m = 3$ gives $n - 4 = -16$, $n = -12$ ✗.

4 Positive factor pairs of 16: $(1, 16), (2, 8), (4, 4), (8, 2), (16, 1)$ - **five** ordered pairs.

why: 'Ordered' means (m, n) and (n, m) count separately: $(5, 20), (6, 12), (8, 8), (12, 6), (20, 5)$.

Where it goes wrong

- Answering 3 by counting unordered pairs when the question says ordered.
- Forgetting $(4, 4)$, the square factor pair.
- Including negative factor pairs without checking they give positive m, n .

Takeaway $\frac{1}{m} + \frac{1}{n} = \frac{1}{k}$ becomes $(m - k)(n - k) = k^2$; then count factor pairs, respecting ordered versus unordered.